# Assignment 3: Alias Analysis

CS6843: Program Analysis

**Submission Deadline:** *Thursday, October 02, 2025, 11:59 PM IST*

## Objective

In this assignment, you will implement a simple Andersen's pointer analysis using LLVM. Your task is to construct constraints, solve them, and query points-to sets to check whether two variables are aliases.

## Instructions

- Implement a basic **Andersen's pointer analysis**.
- For this assignment, assume that each test case contains only a **single function** (i.e., the analysis is **intra-procedural**).
- You **do not** need to worry about constructs which are not present in the provided test cases.
- Supported types: `int`, `int *`, `int **` (no structs or arrays).
- Global variables may be present and should be included in your analysis.
- You may encounter some **type mismatches** in the test cases, but this is acceptable.
- Multiple dereferences in a single statement (e.g., `**p = x`) will **not** be present.
- **All test cases will contain pointers named `a` and `b`.**
- Statements of the form `p = &p` will **not** be present.
- For each test case:
  1. Build the constraints from the program.
  2. Solve the constraints to generate the `pointsTo` sets.
  3. Query the `pointsTo` sets of variables `a` and `b`.
  4. Determine the intersection of `pointsTo[a]` and `pointsTo[b]`.
  5. If the intersection is non-empty, `a` and `b` are aliases; otherwise, they are not.

# Example

**Input C code:**

```c
int main()
{
    int *a;
    int *b;
    int **h;
    int x, y;

    b = &x;
    a = b;
    b = *h;
    h = a;

    return 0;
}
```

**Output:**

```
{x}
```

- The output shows the intersection of `pointsTo[a]` and `pointsTo[b]`.
- A non-empty set indicates that `a` and `b` are aliases.

# Evaluation

- Evaluation is based on correctly identifying aliasing between `a` and `b`.
- A total of **20 test cases** will be used for evaluation:
  - **10 public test cases** are provided to you.
  - **10 hidden test cases** will be used during grading.
- Each passing test case is worth **1 point**.

# Submission Guidelines

- Edit only the file `analysis/pointerAnalysis.cc`.
- Carefully read the `README.md` and all comments provided in the template.
- Rename your `pointerAnalysis.cc` file to `<rollnumber>.cc` and submit it on Moodle.